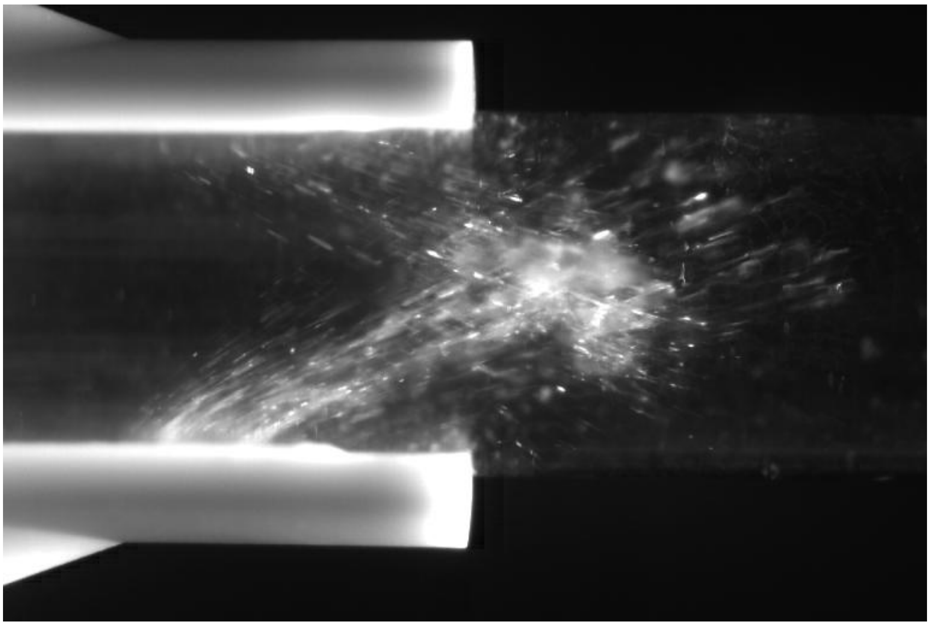
DMT Group 14A
Airflow and Injection

Project Presentation and Test Report



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1.0	19/05/2025	First Draft
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ME3 Design, Make and Test

Executive Summary

With increasing demands to reduce emissions and improve efficiency in internal combustion engines, DMT Project 14A focused on modifying a single-cylinder petrol engine to achieve leaner operation using compression auto-ignition with hydrogen peroxide as a pilot fuel.

The project involved three key development areas: modifying the compression ratio using custom piston crown inserts, designing the fuel delivery system, and designing an air intake system for port side dual-injection of petrol and hydrogen peroxide.

Two piston crown inserts, achieving compression ratios of 11 and 13, were designed and analysed using FEA. The results show a minimum safety factor of 1.9. Thermal stress tests confirm that the usage of boron nitride thermal paste reduces the maximum temperature and ensures minimum safety margin of 174 K against melting, as well as providing confidence that the FEA and manual calculations of safety factors are valid.

The dual-fuel injection system was developed with precise control over injection timing and mass flow rates. CFD simulations were used to determine optimal injection angles, identifying 20° as best possible option. Injecting hydrogen peroxide against closed valve and petrol against the open valve was found to provide the best mixture uniformity while minimising petrol-hydrogen peroxide interaction in the intake port. Finally, the CFD models were experimentally validated with results closely matching the predictions.

The 14A assembly features are, therefore, complete and functional to the require extent set from the project requirements. However, due to delays in the control system, the modified engine could not be thoroughly tested. Hence, future work should focus on integrating the developed product completely with the controls team to perform the superproject testing, determining the optimal hydrogen peroxide concentration and engine compression ratio for successful compression auto-ignition. Additionally, development of direct in-cylinder injection would provide more precise control over combustion timing during compression ignition operation.

1. Overview of the Superproject

In the global push to reduce emissions, electric powertrains are increasingly replacing internal combustion engines (ICE) in transportation. However, in applications where high energy density, operational endurance, and fuel availability are critical, ICEs remain essential. This project aims to address these needs by developing an efficient, low-emission ICE prototype that employs compression auto-ignition (CAI) for a petrol engine, enabling lean operation and improved combustion efficiency.

The core of the project is the modification of a Hyundai IC210X-20 petrol engine to support CAI using a hydrogen peroxide as a pilot fuel to initiate combustion of the air-fuel mixture. The system initiates operation in spark ignition mode before switching to CAI, enabled by a custom-developed engine control unit (ECU) which dynamically manages ignition modes and fuel injection duration using real-time sensor feedback. In addition, the exhaust heat is to be stored in a phase-change material (PCM), which can be redeployed as required.

To successfully achieve the project's objectives, the work was divided among three sub-teams, each focusing on a key technical area, with the overall system shown in Figure 1.

Group A: "Airflow and Injection" - The team was responsible for the design of a dual-injection fuel system capable of delivering a precise mixture of petrol and hydrogen peroxide into the engine cylinder. The team also integrated piston crown inserts to facilitate the transition from spark ignition to CAI operation. This configuration allows for testing a range of fuel mixture concentrations and compression ratios to identify the optimal fuel mixture-compression ratio combination.

Group B: "Thermal Management and Heat Recovery" - This team focused on the development of a waste heat recovery system utilizing PCM to improve the overall thermal efficiency. The team also implemented a dedicated control system to manage PCM charging and discharging cycles and prevent overheating.

Group C: "Mounting, Vibrations, and Engine Control" - The team designed a robust chassis to support the engine and integrate necessary modifications within the testing chamber, ensuring effective vibration damping. Additionally, the team developed ECU hardware and software, integrating various sensors and actuators to enable precise engine control and monitoring.

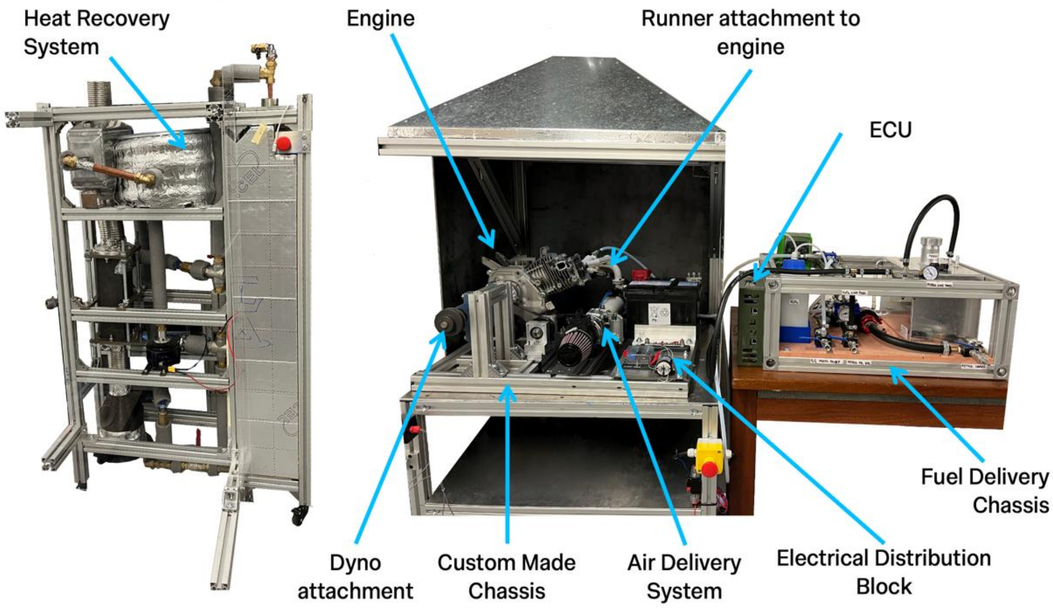

Figure 1: Superproject product overview.

2. Project description

2.1 Overview

The main objective of the Airflow and Injection team is to modify the Hyundai IC210X-20 engine to operate initially in spark-ignition mode to warm up the engine, then switch to CAI. This approach allows for lean air-fuel mixtures, increasing combustion efficiency and reducing harmful NO_X emissions [1]. However, while CAI offers significant advantages on the emission reduction and efficiency front, it also introduces challenges such as knocking and misfiring, which must be addressed to ensure stable and reliable engine operation.

Autoignition of petrol occurs above 553 K but varies with in-cylinder pressure, which can cause knocking or partial burning [2]. Introducing hydrogen peroxide (H_2O_2) as pilot fuel ensures more complete combustion and stable ignition. Studies [1,3,4], typically use a 30% peroxide-water mixture by volume with direct injection near TDC to initiate ignition. Injected masses vary up to 10% by mass of fuel, though some suggest only 1% is needed for CAI [5].

Based on the studies, the system was designed to accommodate H_2O_2 concentrations between 6% and 12% and adjustable injection volumes from 5% to 15% by mass of fuel to test for optimal operation. Three key subsystems were design and manufactured: *(1)* compression ratio modification to achieve conditions suitable for CAI; *(2)* fuel delivery system to store, pressurise, and atomise hydrogen peroxide and petrol; and (3) the air-flow intake system to deliver air-fuel-peroxide mixture into the engine cylinder and ensure a homogeneous mixture is formed at the onset of ignition. Figure 2 below shows the overall system architecture, and the main requirements are in Table 1 afterwards.

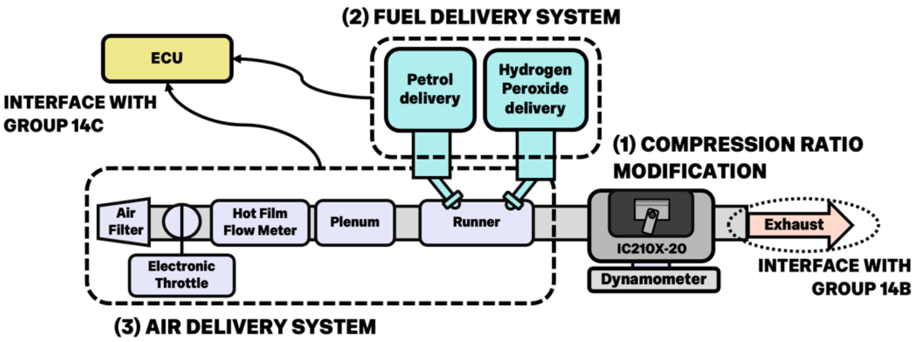

Figure 2: Overall Schematic of the system.

Table 1: Excerpt of the product design specification.

Element	ID	Statement of Criteria
Engine Operation	1.1	Engine running speed is held constant at 1500 RPM
	1.2	Starting operation is with spark ignition which will be then switched to compression auto-ignition with hydrogen peroxide
Compression ratio modification	2.1	Design system for inserts to modify the compression ratio of the engine such that temperature at top dead centre (TDC) is above 553K to guarantee auto-ignition
	2.2	Ensure the modification can withstand forces and temperatures experienced during operation
Fuel delivery system	3.1	Fuel injection timing and dosing to allow proper mixture and atomisation of fuel
	3.2	Peroxide mass flow rate should be in the range of 5% to 15% per petrol mass flow.
	3.3	Fuel system must be resistant to both hydrogen peroxide (max 12%) and petrol
Air flow system	4.1	Deliver well-mixed petrol and hydrogen peroxide air vapor mixture into the engine cylinder
	4.2	Use plenum to increase volumetric efficiency and fit sensor to monitor parameters

2.2 Compression Ratio Modification

Based on PDS point 2.1, the in-cylinder temperature at TDC must reach up to 553 K. This requirement was assessed using a 1D engine simulation with the GT-Power software. Under motored conditions, the maximum and minimum TDC temperatures (accounting for blowby and evaporation variation) were evaluated across various compression ratios (CR). A conservative analysis indicated that a CR of up to 13 is required (for details see Appendix A). This aligns with findings in [2], which suggests a CR of at least 11 is required for CAI. For testing purposes, two piston crown inserts are made with different volumes, so including the original unmodified piston, three CRs of 8.6 (original), 11 and 13 can be achieved. The manufactured product is shown in Figure 3.

Figure 3: Piston with the CR13 crown insert fitted.

For the crowns to be interchangeable for testing, four M3 steel bolts (the largest possible due to space limitations) are used to secure them to the piston. To meet PDS point 2.2, which requires the modification to withstand operational temperatures and forces, basic calculations were performed to find the most likely point of failure. A custom script was used to compute the temperature change using GT-Power heat flux, assuming a lumped capacitance model (with a maximum Biot number of 0.04). The steady-state temperature reaches 830 K, near aluminium's melting point (858 K), hence boron nitride thermal paste (stable to 1123 K) is used to reduce contact resistance. This lowers the maximum the steady-state temperature to 680 K. FEA analysis was performed to first obtain a temperature profile, which was then imposed onto the structure in addition with a 50 N force (corresponding to piston acceleration at 1500 RPM), from which the safety factors were obtained. Figure 4 shows the results at conditions corresponding to full throttle and CR 13 piston crown. The safety factor shows a zoomed in section of the M3 steel bolts.

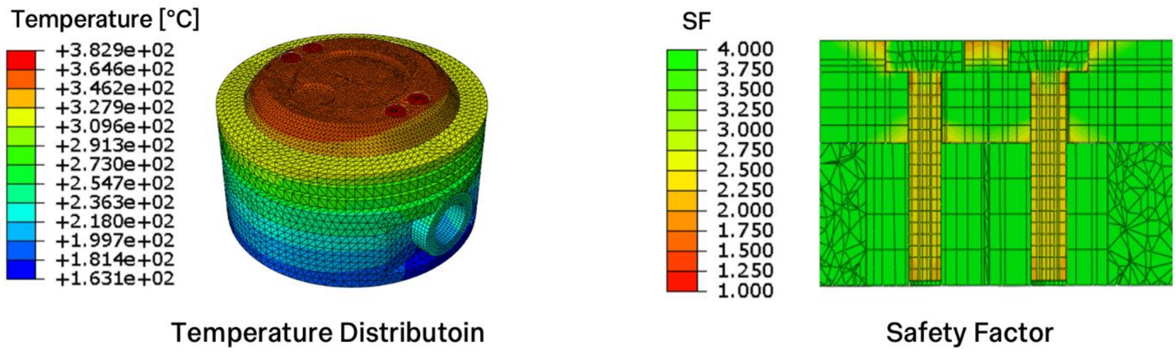

Figure 4: Temperature distribution and safety factor for the compression modification, piston and the bolts.

The lowest safety factor against yielding is 1.9 with an average of 2.3 in the threads. This is consistent to the safety factors calculated manually value of 2.0, considering thermal and tensile stresses for the bolt threads. A preload is applied to the bolts alongside thread locker to prevent loosening by vibrations.

To optimise the likelihood of meeting PDS point 1.2, piston bowls are introduced to improve flame propagation: one is placed in the centre, made as wide as possible to aid in CAI mode, while the second is placed underneath the spark plug to aid spark ignition mode. Sharp corners are also minimised to improve fuel distribution within the cylinder, as well as avoiding very high temperatures created by corners [6]. The impact of these sharp corners was qualitatively evaluated on a motored engine by using CFD, with flow fields shown in Appendix B. As anticipated, the flow reaches the shallow corners more effectively, suggesting that areas of unburned fuel during combustion will be reduced.

2.3 Fuel Delivery System

Precise dosing and timing of injection is required to achieve the correct mass fraction of air-fuel and H_2O_2 mixture. As such the carburettor was replaced with 2 port-side fuel injectors for hydrogen peroxide and petrol, such that PDS point 3.1 is satisfied. Direct injection of hydrogen peroxide was also considered. However, it was later dropped due to high injection pressures and modification of the cylinder head was required. Figure 5 shows the fully assembled fuel delivery system.

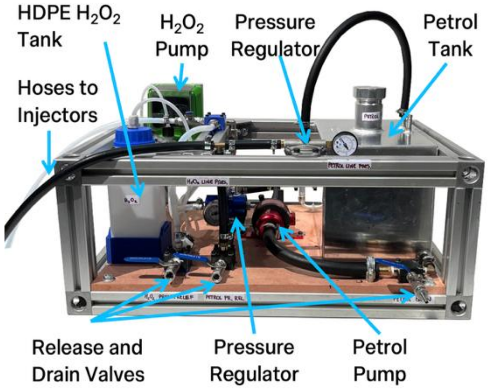

Figure 5: Fuel delivery system.

Injector selection depended mainly on the required fuel mass flow rate, which varies with operating conditions. During spark-ignition operation, the upper bound was determined at full throttle (97% efficiency based on the GT-Power model) under stoichiometric conditions, resulting in a mass flow rate of 1.89×10^{-5} kg/cycle. During compression ignition, the upper and lower bounds of the hydrogen peroxide mass flow rate were identified as 2.83×10^{-6} kg/cycle (full-throttle with 97% volumetric efficiency, stoichiometric operation) and 2.13×10^{-7} kg/cycle (30% volumetric efficiency, air-to-fuel ratio of 20). Refer to Appendix C for detailed calculations. Bosch EV14 CKxT injectors with a flow rate of 347cm^3/min were selected. Their flowrate is large enough for the upper bound requirement, and a fast-closing time for the lower bound.

The petrol system follows the industry standard and comprises of commercial inline pump which can deliver up to 8 bar powered and controlled by Group 14C, a petrol inline filter, a mechanical pressure regulator set to 4 bar, and a quick disconnect fitting for ease of use. The components are fitted together with brass fittings and rubber hoses. Hydrogen peroxide is delivered to the injector with a chemical dosing pump system, capable of delivering 4 L/h at 7 bar with only 0.3 L/h required. The intermittent flow is smoothed and stepped down to 4 bar by a pressure regulator. Satisfying PDS point 3.2, all components of the H_2O_2 delivery system are carefully selected to be compatible with the 12% concentration used during testing. The plumbing components are in PTFE, Silicone, and 316 stainless steel to ensure minimal corrosion and a leak-proof system for safe operation. The P&ID diagrams for both fuel delivery systems are included in Appendix D.

2.4 Airflow and Fuel Injection

The airflow and injection system is responsible for controlling the air intake for the engine, monitoring the intake conditions and providing ports for injection. The assembled system is shown in Figure 6. The air filter, air mass flow rate sensor and the electronic throttle are caried over from the previous project. The plenum has been modified to include a Bosch PST4 pressure and temperature sensor to allow real-time measurement of intake air conditions as required by the PDS.

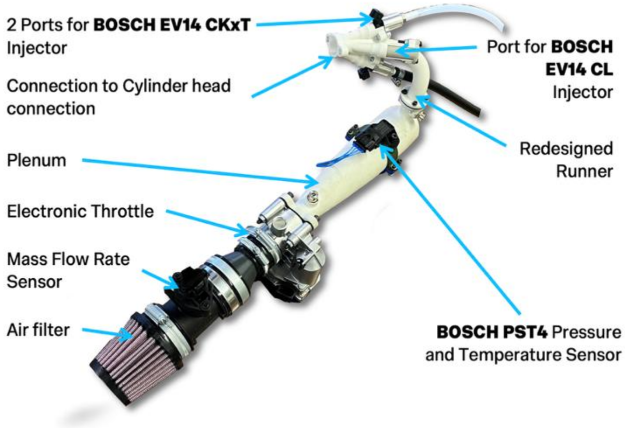

Figure 6: Air Intake Assembly with Installed Injectors.

A new runner was designed to accommodate the 2 injectors. The injection angle was decided by a comparative CFD study where angles of 15°, 20°, 30°, and 40° relative to the streamwise direction were evaluated. A 15° angle minimised fluid film formation and aided in fuel vaporisation by spraying the fuel directly onto the intake valve. However, geometric constraints near the cylinder head prevented this shallow angle for both injectors. Therefore, the second-best option of 20° was used. Originally, the EV14 CL injector was selected for its lower flowrate, enabling precise hydrogen peroxide dosing. However, its short tip caused recirculation zones that trapped small droplets-the ones most likely to fully evaporate. To avoid this, the EV14 CKxT injector with an extended tip was chosen, reducing recirculation but increasing flowrate. As a backup, a third port was added for the EV14 CL injector, which was sealed with a plug for later use if needed. The recirculation zones comparison can be seen in Figure 7. The runner was manufactured by SLS 3D printing from glass fibre reinforced polyamide, which was then sand blasted and treated with sealant to ensure airtight part.

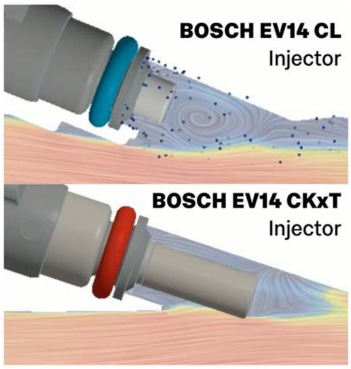

Figure 7: Circulation zone comparison for EV24 CL and EV14 CKxT injector.

Well mixed and uniform distribution of petrol and H_2O_2 is desirable to ensure consistent combustion. To achieve this, a 1D engine simulation (GT-Power) was used to obtain boundary conditions for in-cylinder charge motion analysis. Nine different injection cases were tested, varying between injection against open and closed valves, and injection order, to determine the optimal timing. The most effective strategy identified was hydrogen peroxide injection against closed valve at 0° crank angle (TDC of the compression stroke), followed by petrol injection at 400° crank angle (corresponding to open valve). This method resulted in a more uniform hydrogen peroxide distribution and reduced interaction of both H_2O_2 and petrol in the intake port. Simulations were performed at stoichiometric conditions and lowest H_2O_2 dilute concentrations to ensure the worst-case scenarios are accounted for with the longest injection durations required. Comparative results of the analysis can be seen in Figure 8 which shows H_2O_2 + water vapor distribution for the scenario mentioned above and.

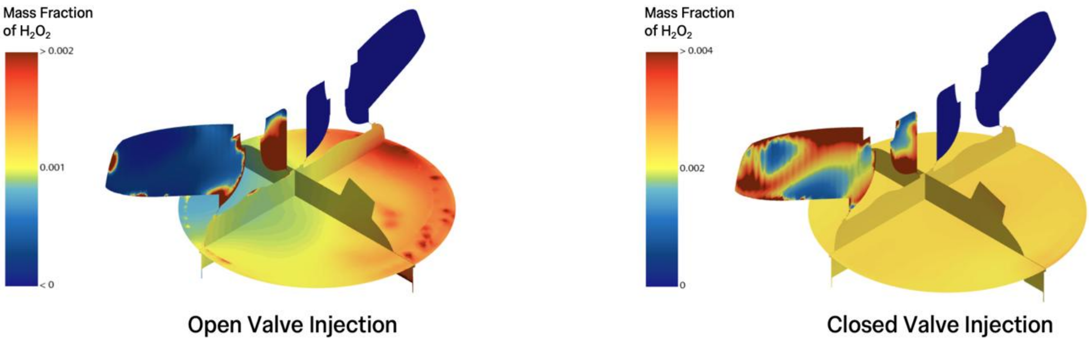

Figure 8: Cylinder X, Y, Z cross section planes for mass fraction of hydrogen peroxide at top dead centre for injection against open valve (left) and optimal timing against closed valve (right).

2.5 Project Budget

The estimated budget for the project was £1,222.25 with the actual cost of £1,455.82. The budget summary is shown in Table 2 below, with the full budget breakdown in Appendix E.

Table 2: Budget Summary by Section.

System	Budgeted (gateway)	Actual	Variation from budgeted
Petrol Delivery	£ 145.99	£ 230.34	£ 84.35
Hydrogen Peroxide Delivery	£ 724.19	£ 777.56	£ 53.37
Air Delivery	£ 199.56	£ 222.24	£ 22.68
Compression Ratio Modification	£ 91.07	£ 51.78	-£ 9.67
Testing	£ 61.45	£ 173.90	£ 82.83
Total	**£ 1,222.25**	**£ 1,455.82**	**£ 233.57**

3. Injection Visualisation Test

3.1 Test Description and Justification

CFD modelling has been extensively used to optimise the design of the fuel injection system. As such, validation is required to ensure the real design will function as intended. This is achieved by visualising injection in a transparent acrylic pipe of the same diameter as the real runner under two *air flow cases*: *(1)* without air cross-flow, which mimics the closed valve injection of H_2O_2 dilute; and *(2)* with air cross-flow, which mimics open valve injection conditions as during the petrol injection. A centerline intake air velocity of approximately 30 m/s was selected to match the 12 g/s average intake mass flow rate of the engine.

Three injection profiles were tested per air flow case: *(A)* top injector at 1 ms alone (corresponding to peroxide injector); *(B)* bottom injector for 5 ms alone (petrol injector); and *(C)* both injectors at the same time for their respective duration. Injections were performed at 4 bar gauge pressure. For safety and ease of handling, water replaced the production fuels (hydrogen peroxide and petrol) during all experiments. The same water-based operating conditions were reproduced in dedicated CFD simulations, enabling a direct comparison between measured and predicted spray characteristics. (Note: From now onwards, test cases are referred to as e.g. *(1A)*, refer to air flow case *(1)* and injection profile *(A)*)

3.2 Methodology and Experimental Setup

The experimental rig, as shown in Figure 9, consists of two of the Bosch EV14 CKxT injectors mounted facing each other at a 20° angle relative to the stream-wise (horizontal) axis, mirroring the orientation used in the real engine. Peroxide and petrol injectors are located on top and bottom respectively. To view the spray without optical distortion from the cylindrical test section, an optically corrected acrylic pipe was designed. Its outer wall geometry was numerically optimised to ensure that parallel incident rays exit without refraction, an approach inspired by [7]. Details of the design and manufacture are outlined in Appendix F.

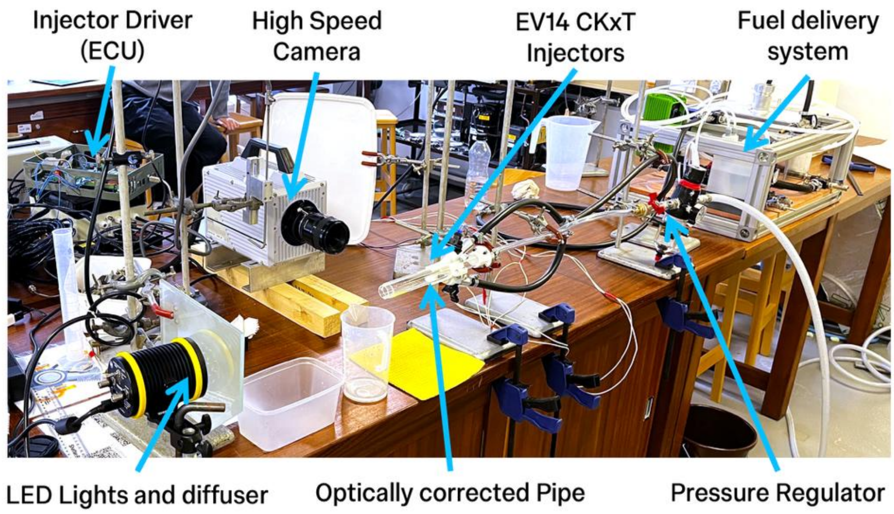

Figure 9: Experimental setup for injection visualisation test (light in position for scattering imaging).

The ECI from group 14C was used as an injector driver to control the injection pulses as required. High-speed camera with 10,000 frames per second was used to record the flow, with the aperture set f/16. The maximum frame size for the given setting of 768 × 512 was sued. Injection and imaging synchronization was achieved through an AVL Engine Timing Unit generating pulses at 10 Hz as a trigger for the ECU and the high-speed camera. Each injection was recorded for 100 frames (test case *(A)*) and 200 frames (test case *(B)* and *(C)*). Overall, 50 injection events were recorded for each test case, resulting in 5,000 - 10,000 frames to be processed.

3.3 Image Post Processing Procedure

The raw data acquired from the camera consists of 12-bit grayscale images capturing the spray events. These images are processed in Python, steps of which are illustrated in Figure 10. The image processing pipeline begins with calculating the average background for each injection event, using the first 15 frames to account for any background variations. This average background is then subtracted pixel-by-pixel from subsequent frames, isolating the spray structure. Next, the image sequence is normalized and denoised using a Gaussian smoothing kernel to reduce high-frequency noise while preserving key features. A thresholding step then converts the processed images into binary masks of the spray. Finally, a median filter is applied to these binary masks to suppress salt-and-pepper noise, ensuring a clean representation of the spray region. This pipeline is adapted from [8,9].

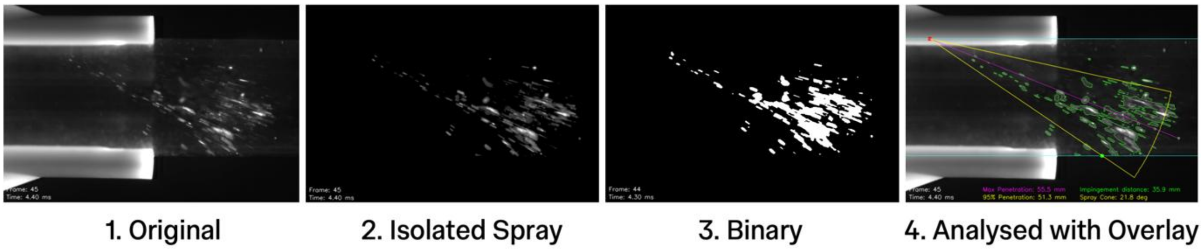

Figure 10: Main steps in image processing pipeline. (Illustrated on example of test case (1A)).

In the test case *(2) with cross flow*, the static-background method failed: lingering wall droplets “twinkled” in the airflow and were counted as spray. Replacing it with OpenCV’s adaptive MOG2 model removed most artefacts, though it slightly eroded the true jet. No single optical setup was optimal: scattering gave the best signal to noise ratio in test case *(1)*, but shadowgraph coped better with twinkling under cross-flow.

From the binary spray images, key parameters, including penetration distance, spray angle, and spray cone, were extracted. Penetration distance was converted from pixels to millimetres using a calibration factor, accounting for non-uniform scaling from optical distortion (1 pixel in y equals 0.82 pixels in x). Spray penetration velocity was estimated using central difference method. The full spray cone was represented by 95th percentile wedge originating at the injector. These parameters were computed for all 50 injection events per test case, from which the mean and standard deviations were computed.

3.4 Results and Discussion

The obtained penetration and velocity curves are presented in Figure 11 and Figure 12, respectively. Error bars show ±1 standard deviation over the injections. In both plots, the time origin corresponds to the trigger pulse; an electronic delay of 0.6 ms is applied before which accounts for the injector opening delay, resulting in injection at 1.5 ms.

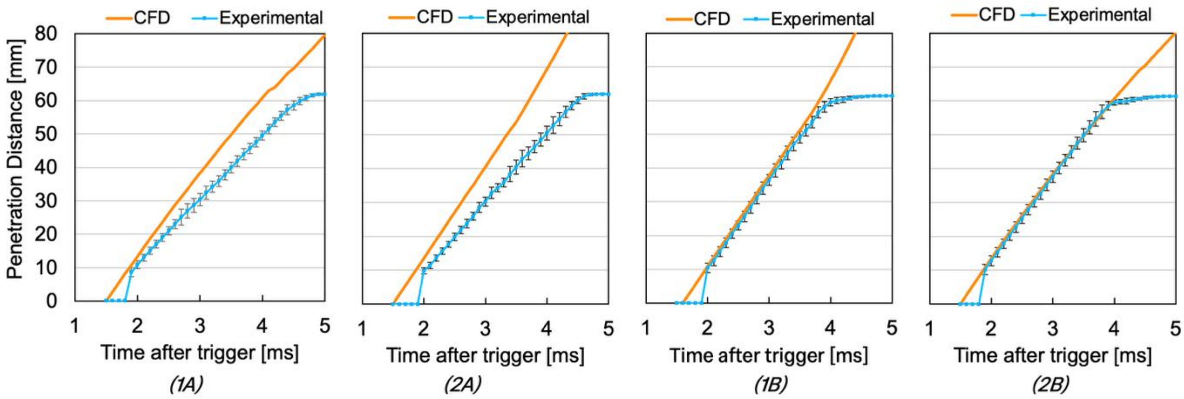

Figure 11: Maximum penetration curves for airflow test cases (1) and (2) with injection cases (A) and (B).

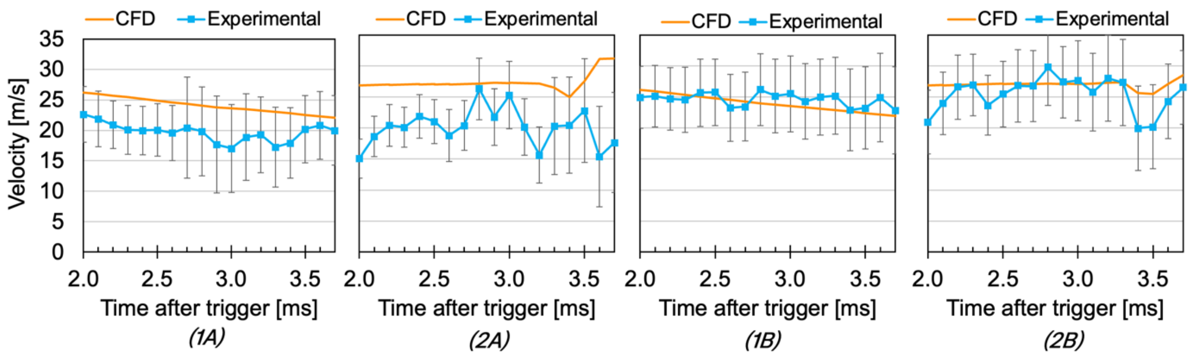

Figure 12: Velocity curves for airflow test cases (1) and (2) with injection cases (A) and (B).

For the bottom-mounted injector (cases *1B* and *2B*) the experimental penetration agrees almost perfectly with the CFD prediction up to about 60 mm, after which the two curves separate because droplets leave the camera's field of view whereas the simulation continues to track them. The top-mounted injector (cases *1A* and *2A*) exhibits the same general trend, but the experimental slope is slightly shallower. A steep initial rise is evident in all experimental traces because the injector tip itself is obstructed, so the first visible droplets are already several millimeters downstream. Over the entire 0-60 mm range the penetration remains slope reduces only slightly, indicating minimal droplet deceleration. This suggests that atomisation of water at 4 bar produces relatively large droplets, as maximum penetration is biased towards the larger ones, whose momentum is not quickly dissipated.

Velocity data in Figure 12 reinforce these observations. For the top injector the initial experimental speed is 22 m/s without cross-flow and 15 m/s with cross-flow, both falling short of the CFD value of ≈ 27 m/s (derived from nozzle area and mass-flow rate). The injector had been used previously, so deposits due to improper flushing after use may have limited needle lift or clogged the nozzle holes and reduced discharge velocity. In contrast, the brand-new bottom injector which matches the CFD well. When cross-flow is present (case *2*) both CFD results display a brief dip followed by an acceleration at around 3.5 ms as the outer most droplets enter the higher-speed core of the pipe flow. This is also briefly observed in the test case *2B*. However, the velocity data is noisy with large standard deviation. And since the imaging window is not large enough it is not possible to judge whether the trend would continue. Further testing with higher temporal resolution and larger imaging window shifted towards the right would provide more conclusive results.

Without cross-flow the spray cone is ~20°, matching the spec. Cross-flow narrows the cone, most strongly in case 1A as it's slower, and a shorter pulse injects less momentum, so the droplets are swept together more. Furthermore, CFD predicts wall impingement at 39 mm (no cross-flow) and 35 mm (flow); experiments agree within results within 1-2 standard deviations, with the furthest experimental impingement for *(1A)* at 44 ± 7 mm. These are shown for all the test cases in the figures below.

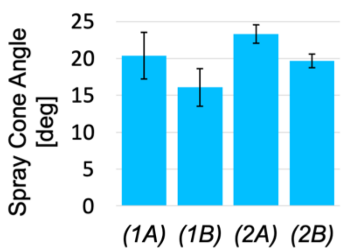

Figure 13: Average spray cone angle.

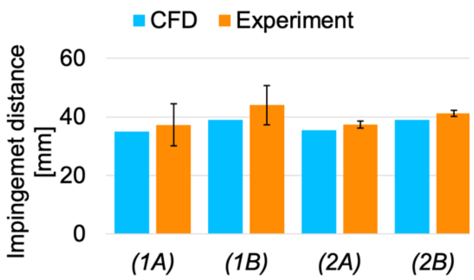

Figure 14: Horizontal Impingement distance per test case.

The dual-plume geometry in case *(C)* prevents use of the automated spray-processing method described in Section 3.3, so Figure 15 provides a qualitative, time-resolved comparison between experiment and CFD for the cross-flow variant (2C).

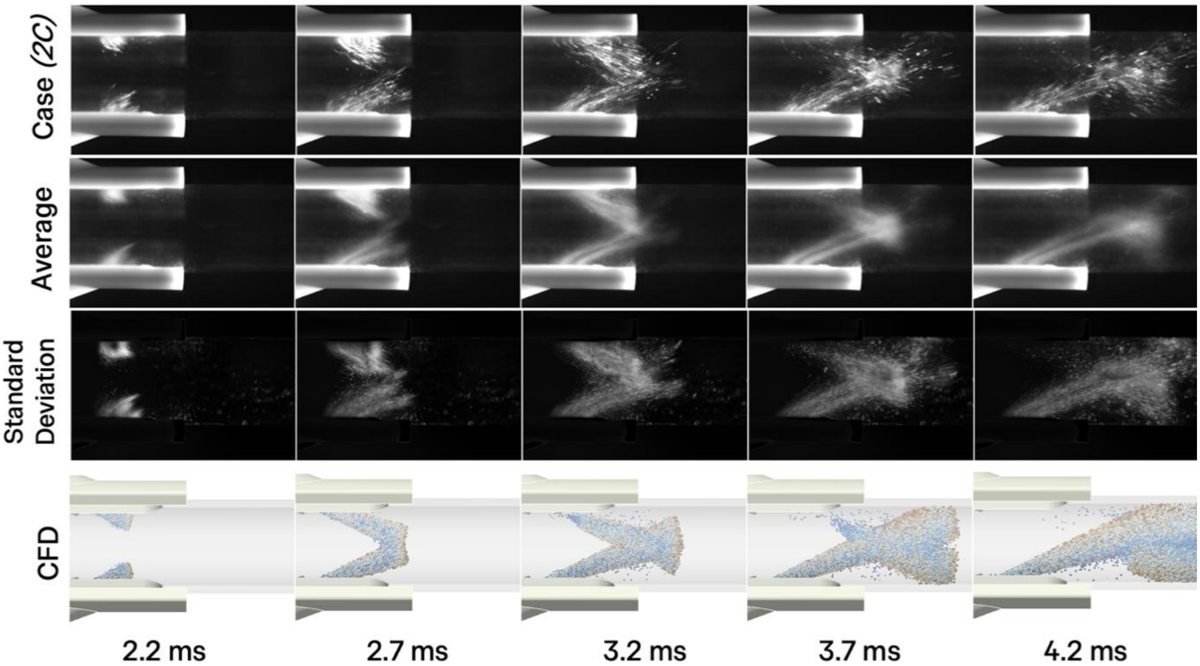

Figure 15: Time evolution of test case (2C) including average, standard deviation and comparison to CFD.

At 2.2 ms, the two rows look nearly identical, but thereafter the lower discharge speed of the top injector becomes evident: CFD plume collision begins at 2.65 ms in the centre of the pipe, where the experimental is delayed to 3.0 ± 0.1 ms. This is most likely caused by the lower injection velocity for the top injector. The CFD also shows a smoother, more uniform plume structure, because each droplet parcel is injected at regular intervals, effectively representing the mean rather than the shot-to-shot fluctuations of the real spray, whereas the standard deviation of the real test case shows greater variability. However, the average spray resembles the CFD suggesting the uniform distribution setting is justified.

3.5 Conclusions and Design Implications

The CFD model accurately predicts the measured spray penetration, velocity and wall impingement. By implication, it can be concluded that the target of injecting the spray directly into the cylinder head is achieved. The main discrepancies between the CFD and experimental results are identified as the lower injection velocity of the top (peroxide) injector and a non-uniform spray. The lower injection velocity and poor atomisation and non-uniform spray can be addressed by increasing the injection pressure.

Several experimental constraints should be addressed in future work. Atomisation was assessed only qualitatively, so droplet-size inputs to the CFD still rely on assumed Rosin-Rammler parameters; direct measurements would refine the predicted in-cylinder vapor air fuel mixture as evaporation is heavily dependent on droplet surface area. Phase Doppler anemometry could be employed to quantify the droplet distribution [8,9]. The optical pipe performed its function well, however, a small curving defect exists near the injection opening. Furthermore, water was used as a surrogate fluid. Both factors introduce uncertainty that could be eliminated by repeating the study with the actual petrol and hydrogen-peroxide blends and a remanufactured optical pipe. Quartz could be used for the pipe, which is more expensive but has better optical properties than acrylic. Lastly, the viewing section should be extended to allow the spray to be observed further downstream.

In conclusion, despite these minor experimental limitations, the CFD model provides a reliable and consistent representation of the spray characteristics, and hence it can be concluded with a certain level of confidence that the peroxide petrol blend within the cylinder would be achieved as required.

4. Piston Crown Thermal Stress Test

4.1 Test Description and Justification

To validate the analysis conducted on the piston modification, a test was designed to simulate relevant operating conditions. The modification is secured to a rig and heated, with strain gauges attached to the bottom face to monitor deformation. Simultaneously, the bottom of the piston is cooled to establish a thermal gradient. From calculations, a thermal gradient of 90 K is expected, and a maximum temperature gradient of 75 K is reached in tests.

The main objective of this test is to verify the analysis made on the piston modification withstanding the forces and temperatures in engine operating conditions. Factors that need to be confirmed include the lumped capacitance assumption, effectiveness of thermal paste, and verification of the FEA.

4.2 Methodology and Experimental Setup

Figure 16 shows the modification with strain gauges secured in the rig, with space for thermocouples and strain gauges to be placed. There are three strain gauges placed at different points with three thermocouples at locations beside them. Two other thermocouples are used. One measures the temperature at the top of the modification, while the other is placed at the exit of the cooling water outlet. The clamp contains channels for water maintained at constant temperature for cooling. A heat lamp is held above the modification as the heat source. Figure 17 shows the overall test set up. The modification is painted black with high temperature resistant paint to increase the absorptivity, maximising the temperature achievable. Exposed wires and hoses are shielded with a glass-fibre sleeve.

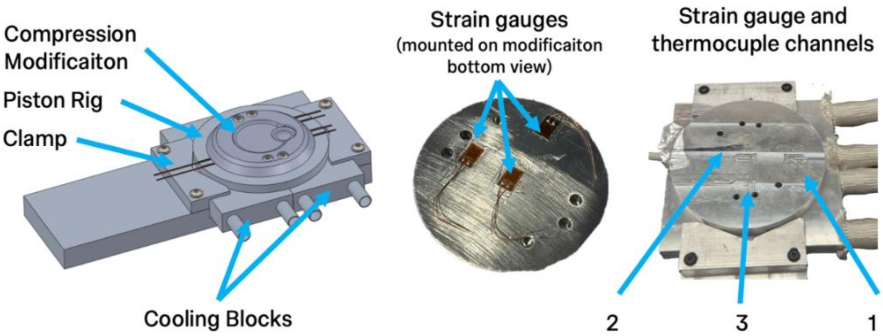

Figure 16: Thermal testing rig, including cooling blocs, piston rig and the compression modification.

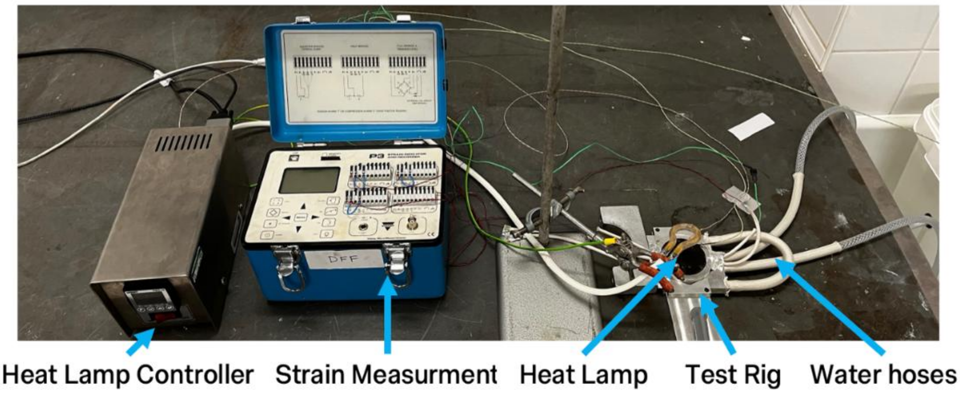

Figure 17: Overall test layout for piston crown thermal test.

Vishay P3 Strain Recorder with Pico Log is used for continuous strain and temperature recording respectively.

The testing process is as follows:

1. Start a timer and start recording readings from thermocouples and strain gauges.
2. At 10 seconds, turn on the heat lamp.
3. At 5 minutes, steady state temperature, turn off the heat lamp.
4. At 8 minutes, stop recording.
5. Repeat the above steps 3 times. Then move the heat lamp closer to the modification, taking note of the height.
6. After 4 different heights tested, repeat the above steps again with thermal paste applied between the modification and piston.

4.3 Results and Discussion

First, the charging and discharging curves are obtained, from which the Biot number is calculated with the following equation:

$$\frac{T(t) - T_i}{T_f - T_i} = e^{-Bi \cdot Fo} \tag{1}$$

Where $T(t)$ is the average temperature between top and bottom surfaces at a given time, t, T_i and T_f are the initial and final temperatures respectively, and Fo and Bi are the Fourier and Biot numbers. The average results of this analysis are shown in in Table 3.

Table 3: Biot numbers for charging and discharging with and without thermal paste

	Charging	Discharging
No Thermal Paste	$(2.7 \pm 0.1) \times 10^{-3}$	$(3.6 \pm 0.4) \times 10^{-3}$
Thermal Paste	$(5.1 \pm 0.4) \times 10^{-3}$	$(9.7 \pm 0.7) \times 10^{-3}$

The theoretical Biot number calculated in Section 2.2 (Bi = 0.04) is approximately an order of magnitude greater than the experimental results discussed earlier. This difference likely arises because the theoretical value is an overestimate: it uses the maximum convective heat transfer coefficient from GT-Power simulations rather than an average. Additionally, the Biot number depends on the specific conditions of the problem, which vary even for the same object - highlighting a limitation of this approach, since exact in-cylinder conditions are challenging to replicate. Nonetheless, the true Biot number is likely between the experimental and calculated values and remains well below 0.1. This suggests that the lumped capacitance model is appropriate and supports the FEA showing uniform piston temperatures with a relatively small temperature difference between top and bottom surfaces of 20 K.

Secondly, the effectiveness of the boron nitride thermal paste is evaluated by varying the heat flux to the piston crown insert. This is achieved by adjusting the height of the heat lamp, which changes the heat flux. For each constant heat flux condition, the maximum temperatures are measured both with and without the thermal paste applied. These temperatures are plotted in Figure 18, where the X-axis shows steady state the temperatures without the paste, and the Y-axis shows the temperatures with the paste.

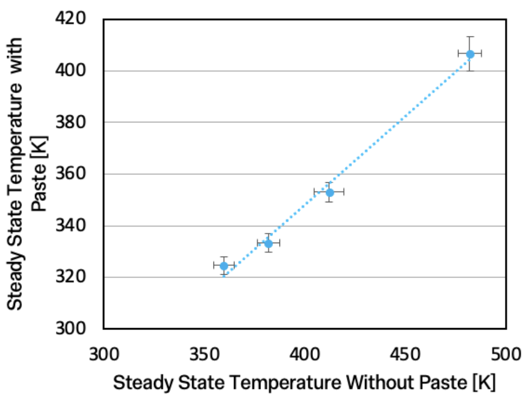

Figure 18: Maximum steady state temperatures with and without thermal paste

The introduction of the thermal paste significantly reduces the maximum temperature, with the reduction increasing linearly with temperature. The best-fit line for this relationship has a gradient of 0.69 ± 0.05. Extrapolating this trend to 830 K (the calculated temperature without paste) yields 643 ± 41 K with paste. The theoretical value of 680 K falls within the error range, though the experiments suggest even lower temperature which is desirable. Hence the piston operates with safety margin of 174 K.

Figure 19 shows the plots of strain (location shown in Figure 16) against temperatures at the corresponding points. Two different temperatures are used: at the strain gauge, and the temperature difference between strain gauge and top of modification.

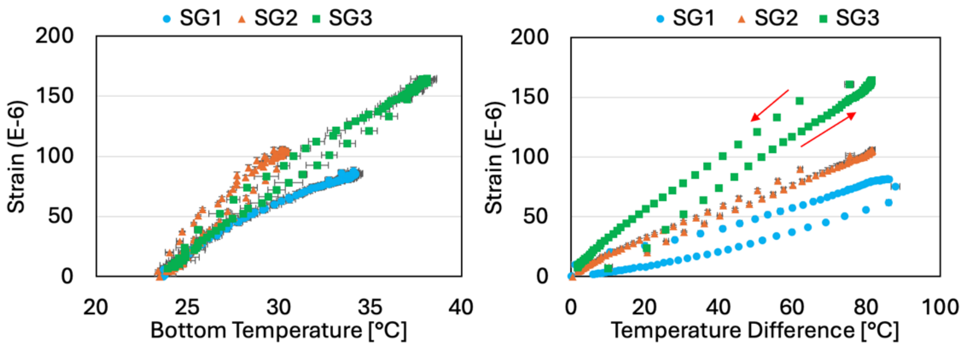

Figure 19: Strain readings against temperature at the strain gauge and the temperature difference.

The strain plotted against temperature difference show hysteresis, while the bottom temperature does not. This is likely because of the time required for heat transfer, with top temperatures changing at a greater rate compared to the bottom surface. The strain gauges may only respond to a change in temperature due to the thermal expansion at that point, with the top temperature being less significant to strain readings. By analysing the average gradients of strain versus bottom temperature, the thermal expansion coefficient is estimated at $(11 \pm 2) \times 10^{-6}$ °C^{-1}. This is significantly lower than the material coefficient for aluminium (2.4×10^{-5}) °C^{-1}, confirming that thermal stresses are significant. Moreover, consistent differences in strain between locations on the piston crown cannot be explained by thermal expansion alone, further supporting the presence of substantial thermal stresses.

An error consistently observed at higher temperatures is that strain readings plateau or even decrease above 120 °C of top piston temperature, instead of increasing linearly. After heating stops, strains quickly return to the expected pattern. This may result from the softening of the epoxy bonding the strain gauges, causing gauges to recover their original length and thus reducing readings. However, the epoxy's rated melting point is 80 °C, which was a temperature not reached at the location of strain gauges. As a result, several strain measurements were discarded.

From Section 2.2, a similar model to the experiment was created in FEA. With the temperature readings from the test, the same conditions are applied to the simulation with the strains obtained from the average of 8 different nodes at each strain gauge location. To verify FEA, especially considering thermal stresses, the strains at points of measurement are compared across a range of steady state temperatures. Figure 20 shows the relation between strain and temperature difference for the test and FEA. Temperature difference is chosen as it is essential for thermal stresses.

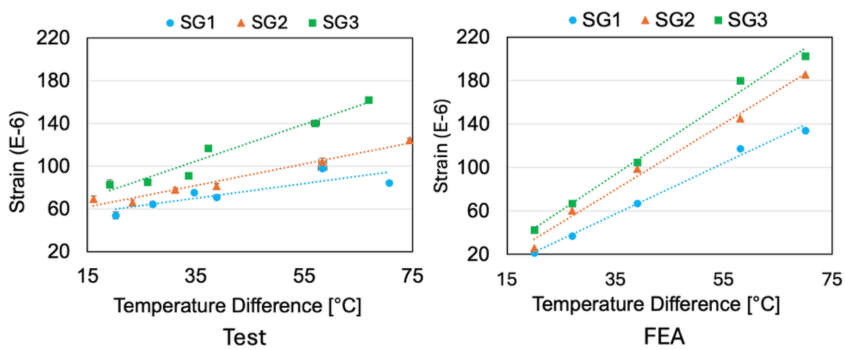

Figure 20: Comparison between FEA model and experimental data

As observed, for the same temperature difference strain gauge 3 has the greatest strains, then 2 and then 1, and the same applies to their gradients as summarised in Table 4. This aligns with expectations as strains follow each gauge's proximity to the bolts: the source of thermal stresses due to materials difference and the fixed joint between bodies of different temperatures.

Table 4: Summary of strain/temperature gradients for experimental data and FEA

	SG 1 Gradient	SG 2 Gradient	SG 3 Gradient
Test	1.7 ± 0.1	1.01 ± 0.09	0.7 ± 0.2
FEA	3.32 ± 0.06	3.05 ± 0.04	2.34 ± 0.04
Difference	48.8%	66.9%	70.1%

The gradients are significantly greater in the FEA compared to test; however, the absolute values are lower in the FEA until a temperature difference of approximately 35 °C Further, the strains of each gauge follow the same pattern, with decreasing values and gradients from gauges 3, 2 and 1. At lower temperature differences, the FEA is a good approximation to reality.

As mentioned previously, the problem of softening epoxy is potentially prevalent at higher test temperatures. This is likely the cause of lower test strains at high temperatures and gradients compared to FEA. This is a clear limitation in the tests and in future, higher temperature epoxy or bonding methods are required.

4.4 Conclusions and Design Implications

For the lumped capacitance test, there is a large disparity between the calculated Biot numbers and the results. The real value cannot be accurately determined as an in-cylinder test must be conducted to do so. Regardless, the lumped capacitance model can be used due to both theoretical and experimental Biot numbers being significantly lower than 0.1. So, the surface temperatures are unlikely to deviate from the 680 K calculated, resulting in a low risk of melting on the top surface of the modification. No redesign is required and a compression ratio of 13 can be used.

The temperature reduction of the modification with the application of boron nitride thermal paste compared to without application was tested for the same set of heat flux. Across all heat inputs, the thermal paste significantly decreases steady state temperature, at linearly increasing amounts as heat flux rises. Results extrapolated to the calculated temperature reduction lay within the bounds of uncertainty. Hence, the thermal paste is proven to be effective and must be applied to all modifications attached to the piston to significantly reduce temperatures and avoid potential yielding or melting.

The steady state strains are investigated across different temperature differences for each strain gauge. Greater strains are experienced at expected points closer to the bolts, but significant deviations are present at higher temperatures (above 120 °C). This is likely due to epoxy softening at higher temperatures leading to these test trials being invalidated. The results from tests are compared to FEA with the same temperature gradients applied. Comparisons suggest that the FEA does quite accurately predict readings from the test at low temperatures and the same relationship between strain gauge readings is observed, however, at higher temperatures, the strains are much lower in experiment resulting in much lower gradients. Epoxy softening is a likely reason for this as well. Nonetheless with the FEA strain/temperature gradients being larger than the experimental values even for lower temperatures where the epoxy is stiff, it can be concluded with confidence that the 1.9 safety factor is likely to be conservative estimate.

5. Outlook for the Developed Project

With an ever-increasing emphasis on efficiency and cleaner energy, the ability to integrate hydrogen peroxide (whose byproducts are simply water and oxygen) into a traditional petrol engine setup has the potential to greatly improve ICE performance and fuel economy in the future, placing the project at the cutting edge of the industry.

Substantially, the prototype developed serves as a starting point for future optimisation processes that can lead to market introduction and, if successfully commercialised, the product will have a strong impact in the industry. As such it is important to understand the steps needed to turn this project into a marketable product, the intellectual property involved in the project, and how to best protect such properties.

To make this prototype commercially viable, it is crucial to study the fuel mixture's life-cycle and usage, as it is the main innovation from standard engines. More specifically, the fuel used must be safe, easy to transport, and compatible with current refuelling infrastructure, and economically sensible to introduce into a multi-billion industry, which may be a concern given hydrogen peroxide's corrosive and potentially unstable properties. In addition, the fuel and air delivery systems should be adapted to the standard geometries of the industry standard chassis, requiring an optimisation process to be carried out on the component selection and placement. The last crucial aspect for commercialisation start-up is the cost reduction process, targeting a wide supplier analysis, manufacturers selection, production lines design and optimisation, and effective use of resources during the production phase.

The designs of the systems produced during the project and the final integration of those components are the main IP involved and the best way to protect them is through patents. However, this presents a set of hurdles that need to be overcome: the patenting of a prototype that results from an academic project in Imperial is a long internal process that requires a clear commercial route which is hard to stipulate at this stage. For this reason, the best approach would be to involve a big player in industry such as Jaguar-Land Rover, which has a collaboration history with the department and a clear competitive interest in the product, to co-operate on the patenting process financially and bureaucracy-wise. The injection system and the compression ratio modification technology can both be patented independently, but if the whole design can be treated as one single system, there may be fewer barriers to the patent application, as independently it would be difficult to make the novelty argument. It would also need to be benchmarked against patents for similar products, as other players in the industry are working on the concretisation of this project (with one of the most promising examples being Caterpillar Inc.'s product described in patent US10119482B1). Therefore, understanding the overlapping features is a crucial process that requires a team of experts to investigate the process.

Additionally, the code developed to control the system can be protected by copyright. However, it may be sensible to consult with the competitors to understand whether open sourcing it could lead to enhancements from other contributors and be beneficial to the overall outcome of the development of the technology, given it is in a very early stage of the commercialisation process. Additional code protected by copyright are the program for spray analysis of high-speed images and the code for the optimisation of the optical pipe's cross section, as these have been developed into small easy to use software framework packages.

6. Conclusions and Recommendations or Further Work

This project focused on the implementing compression auto-ignition using a peroxide-petrol blend in a single-cylinder petrol engine through dual port-side injection. To ensure successful operation, two key subsystems are tested: compression ratio modification and air-fuel delivery.

To withstand operating conditions, calculations and FEA were performed on the modification, which was later verified by a thermal stress test. With the application of boron-nitride thermal paste, a minimum margin of safety of 174 K is achieved from melting. The strains measured in tests follow the FEA in trends, however, due to epoxy softening the results at high temperatures deviate. Despite this, calculations and FEA do predict the modification will withstand operation with a safety factor above 1.9.

The airflow and injection visualisation test validates the CFD spray modeling to a strong degree. By implication this confirms the incoming spray hits the cylinder head near the intake valve aiding in fuel vaporisation reduction of fluid film as intended. However, one of the injectors used was damaged by improper flushing, resulting in reduced flow rate. This needs to be compensated for by increasing pressure or adjusting injection timing. Further work could focus on characterising the droplet size and atomisation of the the fuels directly.

Despite completion of subgroup tests, due to delays in the control group, the overall system test could not be tested in time, preventing integration of the subassemblies and CAI regime operation. Future work should complete the system test to quantify CAI performance for variable parameters of compression ratio, injected volumes of hydrogen peroxide and hydrogen peroxide dilute concentrations. Commercialisation steps include consolidating hardware and software into a single patent, and partnering with industry (e.g. JLR) for validation. Successfully integrated, this concept could reduce emissions of ICEs.

7. References

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Appendices

Appendix A: GT-Power Engine Performance Simulation

To analyse the engine performance and parameters, the engine was simulated in GT-Power. The model simulated can be observed in Figure 21. The engine is first motored at a constant 1500 RPM, with no fuel injection.

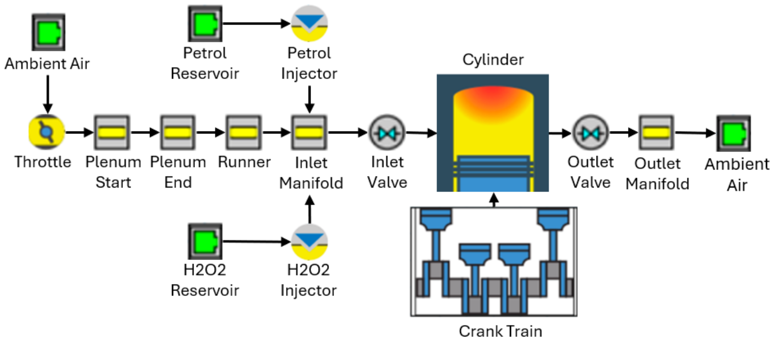

Figure 21: GT-Power simulation component setup

Figure 22 shows the maximum and minimum pressures and temperatures achieved in-cylinder with varying compression ratios. Maximum and minimum conditions are achieved by different angles of throttle, blowby and evaporation. To ensure that CAI can occur, a temperature 553 K is desired which requires a compression ratio of at least 13, to guarantee CAI at all operating conditions, with 11 being on the margin.

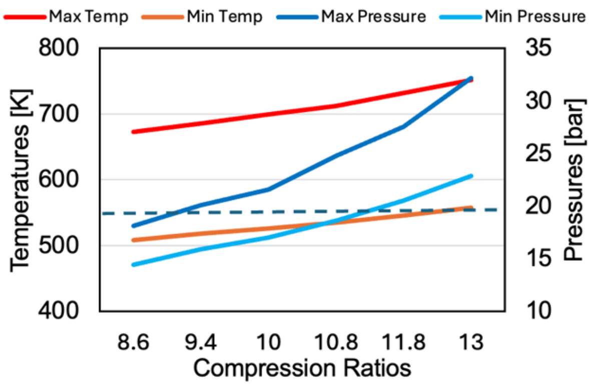

Figure 22: Maximum and minimum pressures and temperatures at varying compression ratios.

Port-side injection of petrol is now introduced with spark ignition to simulate the temperatures the engine will reach with combustion. Figure 23 shows the temperature and heat transfer coefficient with a compression ratio of 13. This information is important in determining the amount of heat transferred to the piston and modification for temperature calculations.

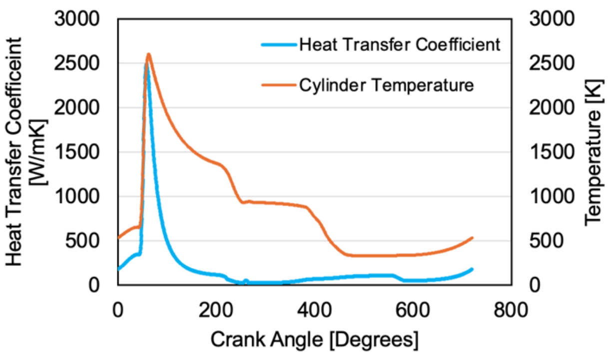

Figure 23: Temperature and heat transfer coefficient at a compression ratio of 13.

Appendix B: In Cylinder flow comparison for Compression Modifications

In-cylinder flow simulations were performed for various compression modification designs to qualitatively assess their performance. A comparison between a simple cylindrical design and the final modification is present in Figure 24 below. Cylinder cross section through the centre of the valve seats is shown at a crank angle of 385°. It can be observed that, in the simple modification, two flow recirculation zones are formed on the left-hand side, in addition to a stagnant flow region at the left bottom corner of the modification. In contrast, in the improved modification, the flow clearly reaches the base of the piston and turns smoothly. This suggests that fuel droplets and vapour introduced during this stage will mix more uniformly with the rest of the chamber. Furthermore, the flame front during spark ignition will flow more smoothly and reach all corners leaving less unburned fuel behind.

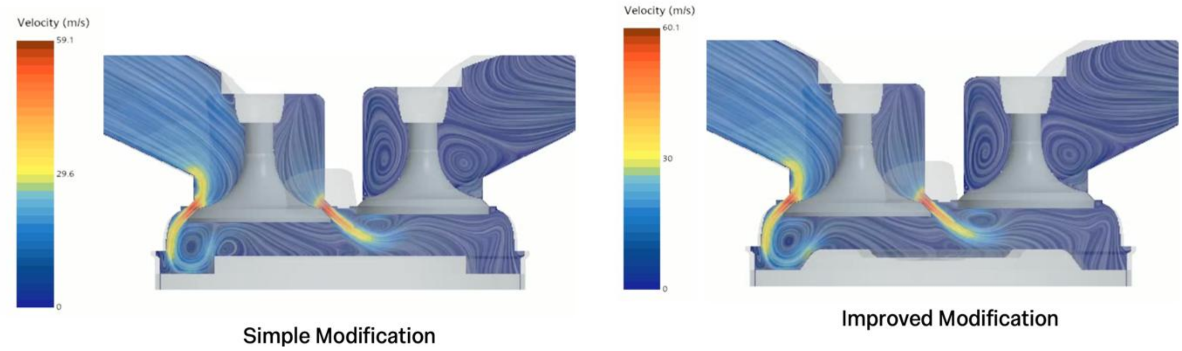

Figure 24: Flow visualisation at 385° crank angle of simple and improved CR11 compression modifications.

Appendix C: Mass Flow Rate Calculations

To select the appropriate injectors, the mass flow rate requirements were identified and then used to determine the boundaries. In Table 5, the maximum and minimum cases are outlined for each of the two fuel components.

Table 5: Flow Rate analysis for fuel components

PETROL FLOW RATE ANALYSIS					
Max			Min		
stoichiometric, volumetric efficiency: 97%, full throttle			*lean, AFR:20, volumetric efficiency: 30%, throttled*		
NAME	VALUE	UNITS	NAME	VALUE	UNITS
$\dot{m}_{average}$	2.356E-04	kg/s	$\dot{m}_{average}$	5.320E-05	kg/s
$\boldsymbol{\dot{m}_{max}}$	**3.393E-03**	**kg/s**	$\dot{m}_{max}$	N/A	kg/s
$m/cycle$	1.885E-05	kg	$m/cycle$	4.256E-06	kg
$\dot{q}_{max}$	4.568E-03	l/s	$\dot{q}_{max}$	N/A	l/s
$q/cycle$	2.538E-05	l	$\boldsymbol{q/cycle}$	**5.729E-06**	**l**
HYDROGEN PEROXIDE FLOW RATE ANALYSIS					
Max			Min		
stoichiometric, volumetric efficiency: 97%, full throttle, 15% mass injection			*lean, AFR:20, volumetric efficiency: 30%, throttled, 5% mass injection*		
NAME	VALUE	UNITS	NAME	VALUE	UNITS
$\dot{m}_{average}$	3.535E-05	kg/s	$\dot{m}_{average}$	2.660E-06	kg/s
$\dot{m}_{max}$	5.090E-04	kg/s	$\dot{m}_{max}$	N/A	kg/s
$m/cycle$	2.828E-06	kg	$m/cycle$	2.128E-07	kg
$\dot{q}_{max}$	4.585E-04	l/s	$\dot{q}_{max}$	N/A	l/s
$q/cycle$	2.547E-06	l	$q/cycle$	1.917E-07	l

Table 6 includes the analysis performed on the petrol system for the selected injector.

Table 6: Injector Analysis for petrol system

	INJECTOR: BOSCH EV14CxT - Petrol Analysis			
	NAME	VALUE	UNITS	NOTES/EXPLANATIONS
injector parameters	$\dot{m}_{N-heptane}$	247	g/min	*g n-heptane/min - defined by injector selected*
	mass flow rate	**4.48E-03**	**kg/s**	***kg gasoline/s***
	t_{delay}	**9.00E-04**	**s**	***injector opening time at 12V***
	$\dot{q}$	6.02E-03	l/s	*l petrol/s*
cycle requirements	cycle time	0.08	s	*defined by PDS required speed of operation*
	V_{needed}	**5.729E-06**	**l**	***from flow rate requirements***
	$t_{open\ targed}$	**9.52E-04**	**s**	*injector time opening to ensure volume needed is delivered at selected flow rate*

Since the injector can deliver a greater mass flow rate than required at maximum, and its minimum opening time is less than the time required for minimum flow rate, the injector is selected. A similar process can be done for the H_2O_2 system.

Appendix D: P&ID Diagrams for Petrol and Peroxide Systems

Figure 25 and Figure 26 below show the P&ID diagrams for the petrol and hydrogen peroxide delivery systems. The symbols used are standard for pipe diagrams.

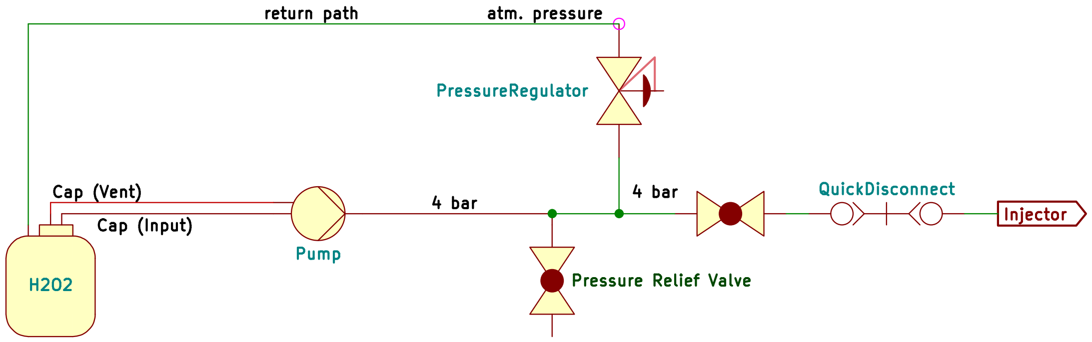

Figure 25: Hydrogen peroxide delivery system diagram.

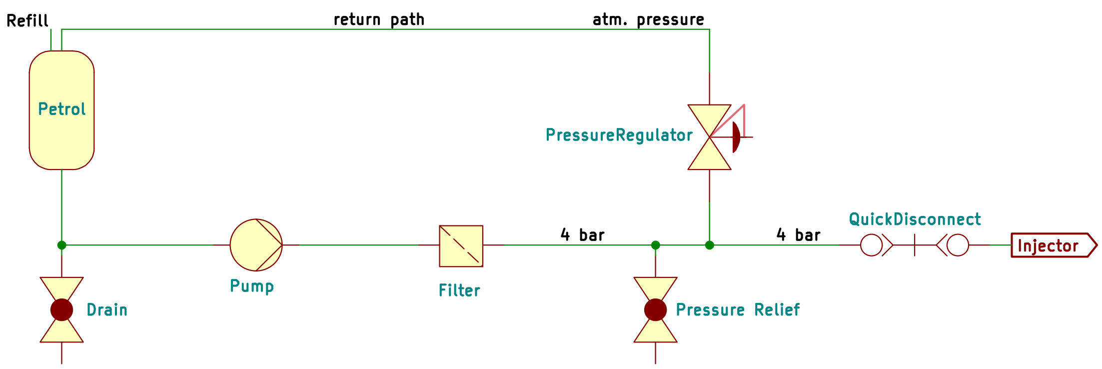

Figure 26: Petrol delivery system diagram.

Appendix E: Budget Breakdown

Table 7: Budget Breakdown by subsystem and supplier.

Description	Spent	Supplier
Air Delivery		
Attachments and Fasteners	£ 16.38	Amazon
Updated heat-resistant runner - 3 injector ports	£ 205.86	Materialise
Hydrogen Peroxide Delivery		
Adjustments for redesigned system	£ 37.45	Amazon
Dosing Pump	£ 299.14	dosingpump.shop
Fasteners	£ 17.34	ME Stores
Fasteners	£ 16.79	Amazon
Fittings	£ 23.71	RS Components
Fittings and Fasteners	£ 97.26	Advanced Fluid Solutions
Fittings for amended design	£ 27.95	Amazon
Hydrogen Peroxide - fuel	£ 16.95	Amazon
Hosetails	£ 9.32	ME Stores
Injector	£ 58.65	Swindon Powertrain
Piping	£ 67.33	Lab Unlimited
Quick-disconnect and piping	£ 87.94	RS
Replacement fittings and fasteners	£ 7.99	ME Stores
Tank	£ 9.74	Amazon
Petrol Delivery		
Hoses and Fittings	£ 39.67	Amazon
Injector	£ 58.65	Swindon Powertrain
Tank cap	£ 15.30	Amazon
AN6 to 8mm hosetail	£ 13.45	RS Components
Brass Male 1/4" BSPT x 5/16" Hose Tail	£ 3.80	ME Stores
Fittings and Fasteners	£ 30.23	Amazon
Fittings	£ 17.01	ME Stores
Refund for 12 mm petrol hose	(£ 6.15)	N/A
Replacement fittings for design amendment	£ 17.39	Amazon
Pump replacement	£ 27.99	Amazon
Quick-disconnect	£ 13.00	RS
Compression Ratio Modification		
Fasteners	£ 16.93	Amazon
Dowel pins	£ 17.00	ME Stores
Aluminium Bar for inserts	£ 2.00	ME Stores
Piston Ring Compressor	£ 15.85	Cromwell
Testing		
Acrylic pipe	£ 8.49	Amazon
Components for thermal test sensors	£ 27.29	RS
N-Type thermocouples for thermal test	£ 12.40	RS
Correct optical pipe for flow visualisation test	£ 114.13	PCB Way
T joint for flow visualisation test	£ 3.99	Amazon
Fasteners	£ 7.60	ME Stores

Appendix F: Design and Manufacture of Optically Corrected Pipe

Flow inside a circular pipe viewed from outside is subject to optical distortion caused by the change in refractive index. This can be modelled with Snell's law as:

$$n_1 sin\theta_1 = n_2 sint\theta_2 \tag{2}$$

Where $n1$ and $n2$ are the refractive indices of on the interface between the two mediums where the ray passes from medium 1 to medium 2, and θ_1 and theta θ_2 are the angles of the incident and refracted ray as measured with respect to surface norma. As a direct consequence, larger refraction occurs near the top and bottom pipe walls, location of particular interest due to spray impingement.

Code was developed to simulate distortion in a circular pipe and correct it by optimizing the outer surface. This was achieved numerically by modelling a pipe with a 20 mm internal diameter and square exterior. 10,000 rays were projected through the pipe in its transverse direction at varying offsets from the centreline, and the outer surface was iteratively modified to ensure each ray exited parallel, eliminating distortion. Simulations results in Figure 27 compare an acrylic pipe with index of refraction 1.49, internal diameter 20 mm and 2 mm wall thickness with the corrected version. Rays (dashed red) pass from the left and get refracted through the pipe cross-sections (solid blue lines). This shows that the optimized pipe maintains parallel rays without distortion. However, a scaling effect remains, requiring correction for quantitative measurements later.

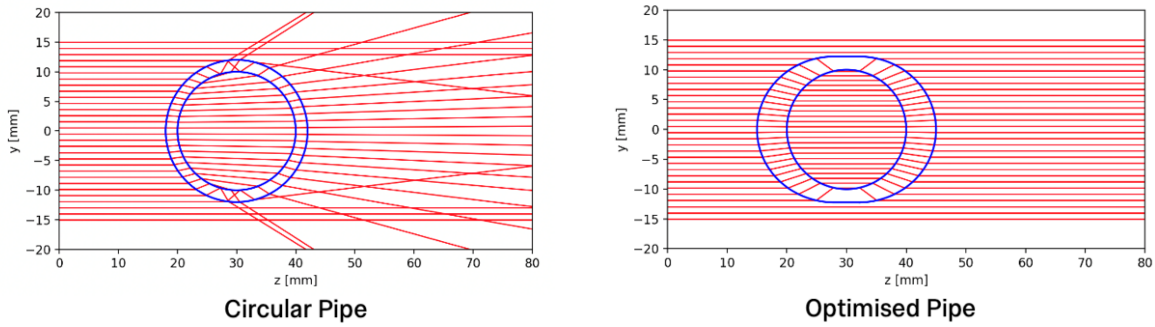

Figure 27: Comparison of ray refraction in circular pipe and optimised pipe

The pipe was manufactured by 5 axis CNC machining from acrylic. To ensure clear optical surface vapor polishing has been utilised. As the pipe has complex outer shape multiple companies were contacted to manufacture with most returning with cost prohibitive quotes (£500 - £1500). PCB Way was the selected manufacturer for price of ~$147 including tax and shipping. The manufactured pipe can be seen in figure below.

Figure 28: Manufactured optical pipe: corrected and uncorrected sections against chessboard background

It can be seen that the optical pipe works as intended but there is slight distortion in the corrected section near the left side. This is location of the injectors. The distortion most likely arises form vapor polishing where due non uniform warping occurred causing distortion. For future the injector openings should be manufactured after vapor polishing to mitigate this. Overall, however the pipe performed its function very well for a low price.